Voluntary Assist-as-Needed Controller for an Ankle Power-Assist Rehabilitation Robot

Renyu Yang, Zhihang Shen, Yueling Lyu, Yu Zhuang, Le Li, and Rong Song, *Senior Member, IEEE*

Abstract*—*Objective: **Although existing assist-as-needed (AAN) controllers have been designed to adapt the robotic assistance to patients' movement performance, they ignore patient's active participation. This study proposed a voluntary AAN (VAAN) controller considering both movement performance and active participation for an ankle rehabilitation robot.** ***Methods:*** **According to the trajectory tracking error of the human-robot cooperation movement, the controller can switch among four working modes, including robot-resist, free, robot-assist, and robot-dominant mode. In order to reflect patients' active participation, the voluntary torque of the ankle joint was estimated by an EMG-driven musculoskeletal model. The control torque in robot-resist, free, and robot-assist mode was determined by the voluntary torque of ankle joint multiplied by an assistance ratio to encourage subjects' active participation, and a stiff torque was provided in robot-dominant mode. The controller was evaluated with 2 healthy subjects and 5 stroke patients on an ankle rehabilitation robot to investigate the clinical impact on the stroke patients.** ***Results:*** **The experiment results showed that as patients' disability level increased, the trajectory tracking error increased and the proportion of human-dominant time and the voluntary torque of ankle joint decreased. Moreover, the results showed that the proposed VAAN controller achieved higher human contribution ratio than that of previous studies.** ***Conclusion:*** **The proposed VAAN controller can adapt the working mode to the movement performance and promote the subjects to participate actively.** ***Significance:*** **Based on its performance, the proposed VAAN controller has potential for use in robot-assisted rehabilitation.**

***Index Terms*—Active participation, ankle rehabilitation robot, human-robot cooperation, performance adaptive, voluntary torque.**

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I. Introduction

More than 80% of stroke patients may suffer from hemiplegic paralysis, which may lead to the impairment in motor function and give a great burden to their family [1], [2]. Rehabilitation training can help patients regain functional use of impaired limbs [3]. Rehabilitation robotics can provide patients with repetitive, sustainable, and customized rehabilitation training to promote motor recovery [4]. Therefore, many kinds of rehabilitation robots have been designed, such as 7-DOF CAREX for upper limb rehabilitation [5], Rutgers for hand rehabilitation [6], ReWalk for lower limbs rehabilitation [7], Robot-assisted AFO for ankle rehabilitation [8], etc.

Control strategies are important in allowing the robot to cater to patients with different levels of motor disabilities. For patients with severe impairments, trajectory tracking control strategies such as the proportion integration differentiation (PID) control strategy can be applied to assist the patient's limb to track a predefined trajectory [9]. Such trajectory tracking control strategies generally provide a consistent guide for patients without consideration of the human-robot interaction, so impedance control strategy was proposed for achieving compliant behavior between the robot and the patient by setting suitable gains of impedance parameters [10]. For patients with less severe impairments, counterbalancing assistance has been developed to support limbs weight in gait training [11]. For patients with slight impairment, a damping force was used in an upper extremity rehabilitation exoskeleton when the movement trajectory of the patient approaches the desired trajectory [12]. The problem with the mentioned control strategies is that they do not take into account the movement performance of patients with different levels of motor disabilities. Assist-as-needed (AAN) controller is proposed to provide the minimal robotic assistance necessary for the patient to complete a motion according to the movement performance [13]. Pehlivan et al. employed an AAN controller and introduced a feedback gain modification algorithm to compute the essential assistive force according to the patient's movement performance [14]. The force-field control strategy was defined to adjust the assistance according to the patient's movement performance [2],

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[15]. Multi-mode control strategies have been certified to realize the AAN controller for rehabilitation robots [16], [17].

Patients' active participation should be encouraged to achieve rapid neuromuscular recovery during rehabilitation training [18], [19], [20]. Most control strategies extract the movement intention based on human-robot physical interaction [21]. Compared with human-robot interaction torque, electromyography (EMG) signals may be a better way to reflect the movement intention because they contain rich information about muscle activity and have a mature and fast non-invasive acquisition technology [22]. Krebs et al. proposed an EMG-triggered controller and the robot assistance was triggered only when the EMG value of the patients reached a certain threshold [23]. But this method cannot achieve EMG-based continuous control, and some studies solved this problem by building linear [24], [25] or non-linear [26] physical mode to correlate the robot assistance and EMG value. EMG-driven Hill-type musculoskeletal models, a non-linear physical mode, were used to reflect the active participation because they have a high accuracy in estimation of voluntary torque according to the physiological and anatomical mechanisms [27].

Although some AAN controllers can provide assistance according to movement performance for patients with different levels of motor disabilities [13], [14], [15], [16], [17], they ignore patients' active participation. EMG signals were previously employed to estimate human voluntary torque and reflect their movement intention, but a fixed assistance ratio was adopted in [19], [28], [29], which cannot achieve the AAN controller. Although EMG signals were used to estimate human voluntary torque in [30], [31], in their studies, AAN controllers were employed to compensate for human torque output according to collaborative tracking performance without promoting subjects' active participation in robot-assist mode and without verifying in patient after stroke. To our knowledge, the combination of realizing the AAN controller and promoting patients' active participation is important but absent in previous studies. In this article, we present a novel voluntary AAN (VAAN) controller, which can both adapt to patients' movement performance and promote their active participation. Firstly, according to the trajectory tracking error of the human-robot cooperation movement, the controller was designed to switch among four working modes (robot-resist, free, robot-assist and robot-dominant mode) to make it adapt to movement performance. Secondly, in order to reflect the subject's active participation, the voluntary torque of ankle joint was estimated by an EMG-driven musculoskeletal model and their difference between the four working modes was determined by different value of the assistance ratio.

II. System Description

The ankle rehabilitation robot is shown in Fig. 1(a) and consists of an aluminum footplate, an AC servo motor (DM1B-045G, Yokogawa, Japan) with an optical encoder, a motor driver (UB1DG3, Yokogawa, Japan), a motor controller (GTS-400-PV(G)-PCI, Googoltech, Hong Kong), a torque sensor (AKC-205, 701st Research Institute of China Aerospace, Science and Technology Corporation, China), a data acquisition device (USB-6341 DAQ card, National Instruments, USA), and

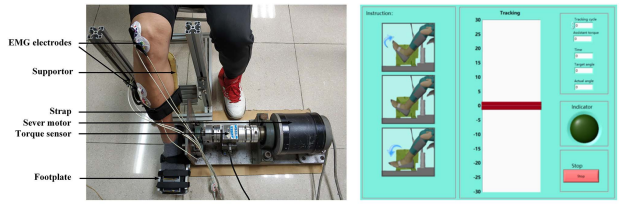


Fig. 1. (a) Photo of the ankle rehabilitation system. (b) User interface.

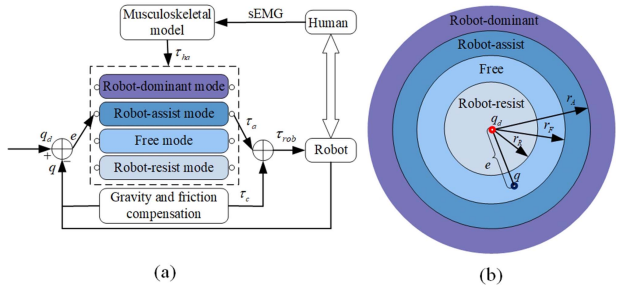


Fig. 2. (a) Control diagram of the ankle rehabilitation system. (b) Control modes.

a personal computer (PC) with software LabVIEW (National Instruments, USA). The two ends of the torque sensor are connected to the footplate and the servo motor, respectively. EMG electrodes and an eight-channel EMG amplifier are used to collect EMG signals of subjects. The robot has a DOF in sagittal plane (plantarflexion/dorsiflexion). Fig. 1(b) shows the user interface, which provides visual feedback to subjects. To ensure the security of each subject during rehabilitation training, two mechanical limits are placed around the footplate to limit the range of angle and the output torque of the servo motor is limited in software program.

III. Control Strategy

The overall control diagram is shown in Fig. 2(a). The output torque of robot, τ_{rob}, consists of τ_a and τ_c. τ_a is control torque, and τ_c is compensation torque for the gravity of the foot plate and friction which used a widespread friction torque model [32]. According to the trajectory tracking error, the controller switches among four working modes, including robot-resist, free, robot-assist, and robot-dominant mode, shown in Fig. 2(b). The first three control modes are human-dominant mode, where τ_a is determined by the voluntary torque of ankle joint τ_{ha} multiplied by an assistance ratio K. The voluntary torque of ankle joint τ_{ha} was estimated by an EMG-driven Hill-type musculoskeletal model, which can detect patient's movement intentions [28]. Whereas in the robot-dominant mode, τ_a is a stiff torque based on the trajectory tracking error.

A. Control Modes

The trajectory tracking error is expressed as

$$e = q_d - q \tag{1}$$

where q_d is desired angle and q is actual angle.

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In order to reflect subject's active participation, τ_a in robot-resist, free and robot-assist mode is the product of the voluntary torque of ankle joint τ_{ha} and the assistance ratio K. The K is negative, 0, and positive values in robot-resist, free and robot-assist mode, respectively. The amplitude of K is determined by movement performance and will be designed later. The value of τ_a in the three control modes is defined as:

$$\tau_a = K\tau_{ha}, \quad |e| \le r_A \tag{2}$$

where r_A is a threshold value.

1) Robot-Resist Mode: When $|e| \le r_R$, the controller switches to the robot-resist mode, where r_R is a threshold value. The controller judges that the subject performs perfectly in ankle movement. A proper resistance can be given to train his/her muscle strength. And then:

$$K = -K_r, \quad |e| \le r_R \tag{3}$$

where K_r is the resistance ratio, which can be adapted to the trajectory tracking error. The adaption of K_r is achieved by a forgetting factor and obtained by the following iterative equation:

$$K_{r_i} = f_g K_{r_{i-1}} + (1 - f_g) K_{r,e} \tag{4}$$

$$K_{r,e} = (r_R - |e|)K_{r,\max}/r_R \tag{5}$$

where K_{r_i} and $K_{r_{i-1}}$ are values of the resistance ratios at iteration k and $k-1$, respectively, $f_g$is the forgetting factor and represents the contribution of the resistance ratio at $k-1$ to that at k, $K_{r,e}$is the contribution of trajectory tracking error to K_{r_i} and $K_{r,\max}$ is the maximum of the resistance ratio. According to (4) and (5), the K_r increases as the trajectory tracking error decreases and vice versa.

2) Free Mode: When $r_R < |e| \le r_F$, the controller switches to the free mode, where r_F is a threshold value. The subject is judged to perform well and can complete the movement by himself/herself. In this case, no intervention of robot will be imposed and K is set to 0.

$$K = 0, \quad r_R < |e| \le r_F \tag{6}$$

3) Robot-Assist Mode: When $r_F < |e| \le r_A$, the controller will switch to the robot-assist mode. In this situation, the subject is judged to have a bad movement performance and needs a proper assistance to complete the movement.

$$K = K_a, \quad r_F < |e| \le r_A \tag{7}$$

where K_a is the assistance ratio, which can be adapted to the trajectory tracking error. The adaption of K_a is achieved by a forgetting factor and obtained by the following iterative equation:

$$K_{a_i} = f_g K_{a_{i-1}} + (1 - f_g) K_{a,e} \tag{8}$$

$$K_{a,e} = (r_A - |e|)K_{a,\max}/(r_A - r_F) \tag{9}$$

where K_{a_i} and $K_{a_{i-1}}$ are values of the assistance ratios at iteration k and $k-1$, respectively, $K_{a,e}$ is the trajectory tracking error's contribution to K_{a_i} and $K_{a,\max}$ is the maximum of the assistance ratio. According to (8) and (9), the K_a will increase if the trajectory tracking error increases and vice versa.

4) Robot-Dominant Mode: When $|e| > r_A$, the trajectory tracking error becomes very large and the controller will switch to the robot-dominant mode. In this case, the movement performance of the subject is judged abnormal and a stiff torque is provided to ensure the safety of subjects:

$$\tau_a = K\left(|e| - r_A\right), \quad |e| > r_A \tag{10}$$

where K is set to 0.5.

B. Hill-Type Musculoskeletal Model

In this study, a Hill-type musculoskeletal model is applied to estimated real-time ankle joint voluntary torque of the subject during rehabilitation training. The inputs of the model are EMG signals and the ankle joint angle of the subject, and the output is the voluntary torque of the ankle joint. The model consists of four parts: muscle activation model, musculoskeletal geometry model, Hill-based muscle–tendon model, and dynamic model of the ankle joint [33].

1) In the muscle activation model, first, the EMG signals are amplified and normalized using the maximum voluntary contraction (MVC) level. And then a full-wave rectification and a 2 Hz bandpass second-order Butterworth filter are utilized to obtain the envelope of processed EMG signals $e(t)$ Based on $e(t)$, the neural activation $u(t)$ could be expressed as follows [34]:

$$u(t) = 0.9486e(t-d) + 0.052u(t-1) - 0.000627u(t-2) \tag{11}$$

where d, the electromechanical delay (EMD), is set as 80 ms [35]. The muscle activation $\alpha(t)$ can be obtained by the following model referring to [36]:

$$\begin{cases} \alpha(t) = a_1 \ln(a_2 u(t) + 1), 0 \le u(t) < u_0 \\ \alpha(t) = a_3 u(t) + a_4, \ u_0 \le u(t) < 1 \end{cases} \tag{12}$$

where $(u_0, \ a_0)$ is the node point of the curve, and could be expressed as follow:

$$\begin{cases} u_0 = 0.3085 - A\cos 45^\circ \\ \alpha_0 = 0.3085 + A\sin 45^\circ \end{cases} \tag{13}$$

The coefficients $a_1, \ a_2, a_3, a_4$ in (12) are determined by A during model calibration referring to [19].

2) In the musculoskeletal geometry model, l_{mt}, the length of muscle-tendon unit and r_{mt}, torque arm of muscle-tendon unit is calculated using the following equations [37]:

$$l_{mt} = \sqrt{\left[\left|\overrightarrow{OA}\right|^2 + \left|\overrightarrow{OB}\right|^2 - 2\left|\overrightarrow{OA}\right| * \left|\overrightarrow{OB}\right| * \cos(q)\right]} \tag{14}$$

$$r_{mt} = \partial l_{mt}/\partial q \tag{15}$$

where $|\overrightarrow{OA}|$ denotes the length between the muscle–tendon origin and the respective joint center, and $|\overrightarrow{OB}|$ denotes the length between the muscle–tendon insertion and the respective joint center. q represents the ankle angle.

3) The Hill-based muscle–tendon model is made up of a contractile element (CE), a passive element (PE), and the tendons. The force generated by muscle-tendon unit, F_{mt}, can be obtained

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TABLE I
BASIC INFORMATION OF THE SUBJECTS

SUBJECTS	GENDER	AGE (YEAR)	HEIGHT (M)	WEIGHT (KG)	HEMIPLEGIC SIDE	FMA-LE SCORE
H1	male	24	1.73	65	right	32
H2	female	24	1.66	62	right	32
S1	male	40	1.67	63	right	29
S2	male	45	1.62	60	left	26
S3	male	50	1.61	62	right	20
S4	male	70	1.60	52	left	18
S5	male	53	1.65	76	right	16

FMA-LE score：the lower-extremity Fugl-Meyer Assessment score (0 = total impairment, 32 = no impairment)

using the following equation:

$$F_{mt} = (F_{ce} + F_{pe}) \cos\phi \tag{16}$$

where F_{ce} and F_{pe} denote the force generated by CE and PE, respectively. ϕ represents the pennation angle. F_{ce} is related to $\alpha(t)$.

4) The voluntary torque, τ_{ha}, can be obtained in the dynamic model of the ankle joint by the following equation [38]:

$$\tau_{ha} = \sum (r_{mt,\,i} \times F_{mt,i}) \tag{17}$$

where $r_{mt,\,i}$ and $F_{mt,\,i}$ are the torque arm and force of the i th muscle-tendon unit, respectively. τ_{ha} is either dorsiflexion torque or plantarflexion torque, which depends on the desired trajectory. And the direction of dorsiflexion is defined positive

$$\tau_{ha} = \begin{cases} \tau_{\text{dorsiflexion}}, & \dot{q} \geq 0 \\ \tau_{\text{plantarflexion}}, & \dot{q} < 0 \end{cases} \tag{18}$$

For calibration, the undetermined parameters in Hill-type musculoskeletal model, e.g., the shape coefficient A, the tendon slack length, $l_{st}(m)$, and the maximum isometric muscle force, are determined by minimizing (19) in MVC experiment [38].

$$Er = \frac{1}{s} \sum_{i=0}^{s} \left(\tau_{ha}(i) - \tau_m(i)\right)^2 \tag{19}$$

where $\tau_m(i)$ and s are the actual joint torque measured by the torque sensor in the i th sampling and the total number of samples in this MVC experiment. The rest of undetermined parameters ($|\overrightarrow{OA}|$, $|\overrightarrow{OB}|$) are set using OpenSim (National Institutes of Health for Biomedical Computation, Stanford, USA) as references. For each subject, a calibration on musculoskeletal model is carried out to obtain calibrated model parameters before experiments [19].

IV. EXPERIMENTS AND EVALUATION PARAMETERS

A. Experiments

To verify the performance of the proposed control strategy, experiments were performed on two healthy subjects (H1 and H2) with no history of neurological impairment and five stroke patients (S1 to S5). The selection criteria of the stroke patients for inclusion included: *1)* patients were diagnosed with hemiparesis due to a single unilateral lesion of the brain; *2)* the voluntary EMG signals of the affected tibialis anterior (TA) and gastrocnemius (GAS) should be measurable with a relatively small amount of contamination of involuntary EMG; and *3)* patients should not have visuospatial, cognitive or attention deficits. Basic information of these subjects was listed in Table I. The Fugl-Meyer Assessment score for Lower-Extremity (FMA-LE, 0 = total impairment, 32 = no impairment) was used and the FMA-LE score of S1 to S5 descended. This study was approved by the Ethics Committee of the Guangdong Work Injury Rehabilitation Center. All procedures were conducted according to the Declaration of Helsinki and all participants provided written consent prior to testing.

TABLE II
CALIBRATED PARAMETERS OF EMG-DRIVEN HILL TYPE MUSCULOSKELETAL MODELS FOR EACH SUBJECT

SUBJECTS	TA			GAS		
	A	$l_{st}(m)$	$F_{mx}(N)$	A	$l_{st}(m)$	$F_{mx}(N)$
H1	0.06	0.2346	638.44	0.06	0.3850	169.14
H2	0.06	0.2321	394.11	0.06	0.5984	120.48
S1	0.06	0.2346	566.67	0.06	0.3850	101.58
S2	0.06	0.2346	299.94	0.06	0.3850	100.00
S3	0.06	0.2389	233.33	0.06	0.2326	28.77
S4	0.06	0.2346	80.00	0.06	0.3850	50.00
S5	0.06	0.2300	152.67	0.06	0.3850	119.75

A: non-linear shape factor in the EMG-to-activation model; l_{st}: tendon slack length; F_{mx}: Maximum isometric muscle force

The experiments mainly consisted of three parts: the MVC experiments, tuning of the control parameters, and the human-robot cooperation movement experiments. First, the subject should sit on a chair in a comfortable way. His/her hemiplegic foot was fixed to the footplate using straps, and the footplate was perpendicular to the ground. Then EMG electrodes were attached to the TA and GAS of the subject. There was a monitor with a user interface in front of the subject showing the desired and actual ankle joint angle, shown in Fig. 1(b).

1) The MVC experiments were carried out to calibrate the parameters of EMG-driven musculoskeletal model. The subjects were instructed to activate their TA and GAS muscle as much as possible with their ankle joint fixed at 90° and 120° of ankle dorsiflexion and plantarflexion, respectively. The calibration at each angle was repeated three times with the duration of 5 s for each trial, and in order to prevent muscle fatigue, there was also a 1-minute break between two trials. The torque and the EMG

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TABLE III
CONTROL PARAMETERS

PARAMETERS	VALUE
r_R	3°
r_F	5°
r_A	9°
$K_{r,max}$	0.1
$K_{a,max}$	0.5

signals were captured simultaneously during the experiments and stored to calibrate in a MATLAB program using (19) [28]. The results of calibration were listed in Table II.

2) The division of working area and assistance ratio K should be designed appropriately to cater for subjects with different levels of disabilities. The designed controller is supposed to provide less robot assistance or even resistance for patients with mild impairment and provide more assistance for patients with severe impairment. After referring to studies [12], [13], [14] and test trials on H1, the control parameters were adjusted and shown in Table III.

3) In the human-robot cooperation movement experiments, the subject was asked to try to follow the desired angle using his/her ankle. The desired trajectory was a sinusoidal trajectory with an amplitude of 25° and a frequency of 0.1 Hz for 30 seconds. Each subject was asked to perform the movement three times and rest for 30 seconds between each movement. The angle, torque, and EMG signals were collected by the DAQ card with a sampling frequency of 1000Hz.

B. Evaluation Parameters

The root mean square error (RMSE) was used to evaluate the subjects' trajectory tracking performance:

$$RMSE = \sqrt{\sum_{i=1}^{N} e^2(i)/N} \tag{20}$$

where $e(i)$ and N represented the trajectory tracking error at the i th sampling instant and the total sample number in the human-robot cooperation movement experiments, respectively.

The proportion of human-dominant time was given by:

$$P_{hd} = t_{hd}/t_{total} \times 100\% \tag{21}$$

where t_{hd} was the time when the controller was in human-dominant mode and t_{total} was the total experiment time.

The root mean square (RMS) of τ_{ha} was used to quantify the human voluntary torque output in four working modes, respectively. The metric of each working mode was defined as follows:

$$RM\,S_{\tau_{ha}} = \sqrt{\sum_{i=1}^{w_j} \tau_{ha}^2(i)/w_j} \tag{22}$$

where w_j $(j = 1, 2, 3, 4)$ denoted the total number of data points in robot-resist, free, robot-assist, and robot-dominant mode, respectively, and $\tau_{ha}(i)$ denoted the i th data point of τ_{ha} in the corresponding working mode. In a similar way, the RMS of τ_{rob} was used to quantify the robot torque output in four working modes, respectively. τ_{rob} consisted of τ_a and τ_c, in robot-resist mode $\tau_a$was always in the opposite direction as τ_{ha}, whereas τ_c in the same direction as τ_{ha}. Considering the direction of torque, the RMS of τ_{rob} in robot-resist mode was defended as

$$RM\,S_{\tau_{rob}} = -\sqrt{\sum_{i=1}^{w_1} \tau_a^2(i)/w_1} + \sqrt{\sum_{i=1}^{w_1} \tau_c^2(i)/w_1} \tag{23}$$

Whereas $\tau_a$was always in the same direction as τ_{ha} in free, robot-assist, and robot-dominant mode, thus the RMS of τ_{rob} in these working modes was defended as

$$RM\,S_{\tau_{rob}} = \sqrt{\sum_{i=1}^{w_j} \tau_a^2(i)/w_j} + \sqrt{\sum_{i=1}^{w_j} \tau_c^2(i)/w_j} \tag{24}$$

where $\tau_a(i)$ and $\tau_c(i)$ denoted the i th data point of τ_a and τ_c in the corresponding working mode, respectively; and $j = 2,\ 3, 4$.

The human contribution ratio in four working modes was used to evaluate patient's active participation [30], [31], [39]:

$$R_h = RMS_{\tau_{ha}}/(RMS_{\tau_{ha}} + RMS_{\tau_{rob}}) \tag{25}$$

R_h denoted the human contribution in percentage of a full contribution to the movement: a higher R_h indicated that subjects maintained higher level of active participation during the movement. and vice versa [39].

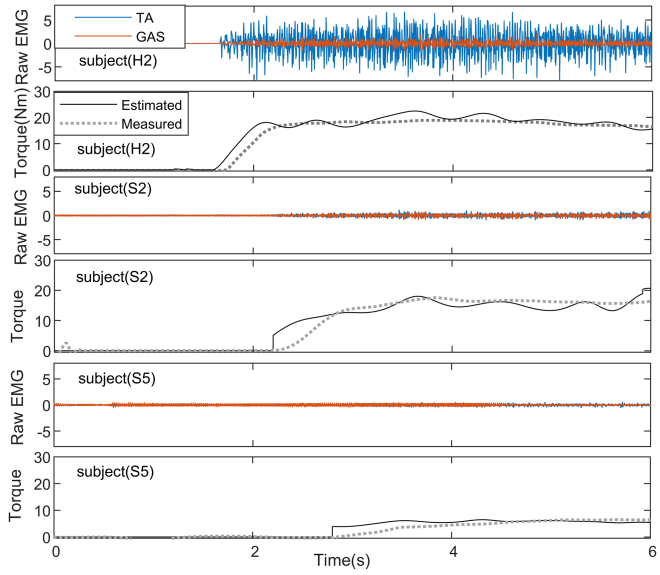


Fig. 3. The raw EMG signals and the estimated MVC ankle torque using EMG-driven musculoskeletal model for H2, S2 and S5 when the ankle joint was positioned at 90.

V. RESULTS

Fig. 3 demonstrated that the accuracy of human torque estimation for H2, S2, and S5 when the ankle joint was positioned at 90° in the MVC experiments and the estimation error of each participant's torque was at a low level. The trajectory tracking error, the K, τ_{rob} and τ_{ha} of H1, S2, and S5 during the first 10 seconds were shown in Fig. 4, where the positive

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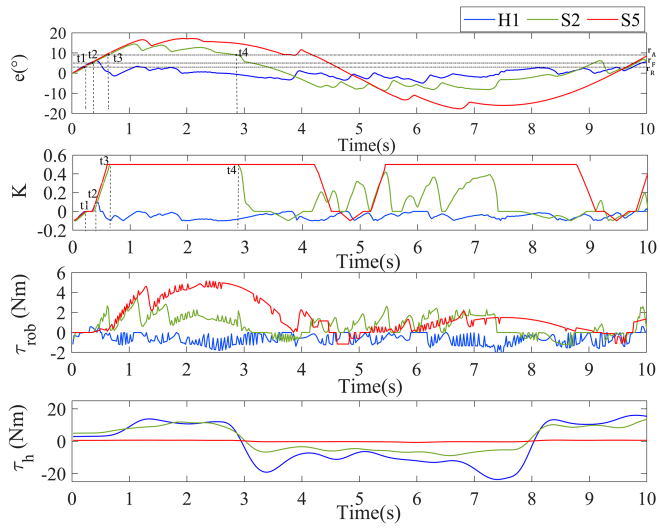


Fig. 4. The trajectory tracking error, K, τ_{rob} and τ_{ha} of H1, S2 and S5 during the first 10 seconds. The sign of τ_{rob} is defined according to type of τ_{rob}(assistance is positive, whereas resistance is negative). The sign of τ_{ha} is defined according to $\dot{q}$.

τ_{rob} and K meant assistant torque, and the negative τ_{rob} and K meant resistant torque. According to the trajectory tracking error, the controller switched to the corresponding working mode. Take S2 as an example, during $t \in [t_0, t_1)$, $|e| \leq r_R$, K and τ_{rob}were negative and the controller were in robot-resist mode; during $t \in [t_1, t_2)$, $r_R < |e| \leq r_F$, K and τ_{rob} were 0, which meant the controller switched to the free mode; during $t \in [t_2, t_3)$, $r_F < |e| \leq r_A$, K and τ_{rob} were positive and the controller switched to the robot-assist mode; during $t \in [t_3, t_4)$, $|e| > r_A$, the controller switched to the robot-dominant mode. The controller provided proper working modes for subjects with different trajectory tracking error. Fig. 4 reflected that the error of H1 was the smallest and the controller mostly switched to the robot-resist mode. The error of S2 was larger than that of H1 and the controller was mostly in the human-dominant mode. The error of S5 was the largest and the controller mainly switched to the robot-dominant mode.

Besides the working modes, the output torque of the robot depends on the K and τ_{ha}. It can be seen that K can change adaptively based on the trajectory tracking error in Fig. 4. Specifically, the K_r and K_a increased with increasing trajectory tracking error. In addition, during $t \in [t_2, t_3)$, the K_a of S2 and S5 were similar, while the assistant torque of S2 was larger than that of S5. The reason is that S2 had larger τ_{ha}, which will be shown in Figs. 4 and 7.

The trajectory tracking error of subjects was shown in Fig. 5 and Table IV. It can be seen that healthy subjects (H1 and H2) tracked the desired trajectory well and the RMSEs of trajectory tracking error were less than 3°, which were smaller than those of patients. Among patients, S1 and S2 had better performance and the RMSEs were less than 5.6°. While the other patients (S3, S4 and S5) had difficulty completing the movement task, resulting in large error with RMSEs more than 10°. As can be observed, the trajectory tracking error of subjects increased as the FMA-LE score decreased.

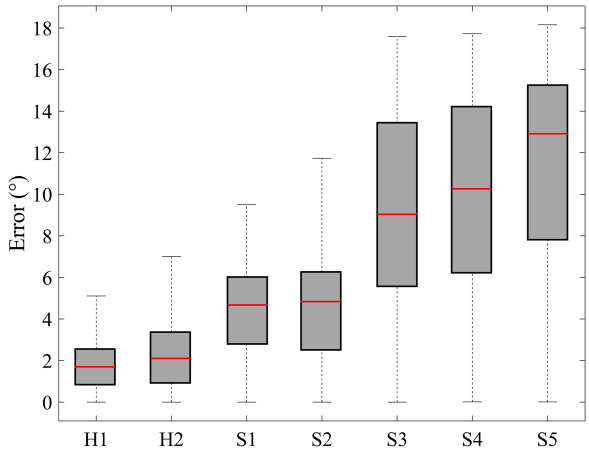


Fig. 5. The box plots show the median (red line), inter-quartile range (box), and minimum and maximum values (tails) of the amplitude of trajectory tracking error from 2 healthy subjects and 5 stroke survivors.

TABLE IV
STATISTICAL RESULTS FOR THE HUMAN-ROBOT COOPERATION MOVEMENT EXPERIMENTS

SUBJECT	RMSE(°)	P_{hd} (%)	$RMS_{\tau_{ha}}$ (Nm)
H1	2.18	100	13.14
H2	2.82	100	4.36
S1	4.95	97.67	9.86
S2	5.57	92.54	8.39
S3	10.26	49.97	2.97
S4	11.08	37.01	2.05
S5	12.26	30.38	0.55

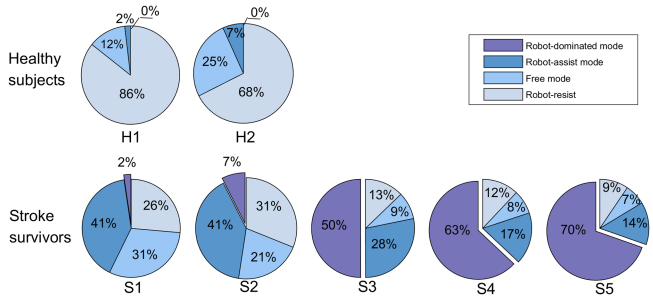


Fig. 6. Proportion of time of each working mode from 2 healthy subjects and 5 stroke survivors.

Fig. 6 and Table IV showed the proportion of human-dominant time P_{hd} of subjects. As can be seen, the proportion of resist-mode time of healthy subjects was more than 60%, which meant they completed the desired movement against the robot resistance in most time. The P_{hd} of healthy subjects was 100% and the controller was always in human-dominated mode. The proportion of resist and free modes time of S1 and S2 were more than 50% and they completed the movement without robot assistance. The P_{hd} of S1 and S2 were more than 90% while the P_{hd} of the other patients was less than 50%. The controller mostly switched to robot-dominant mode and provided assistance for

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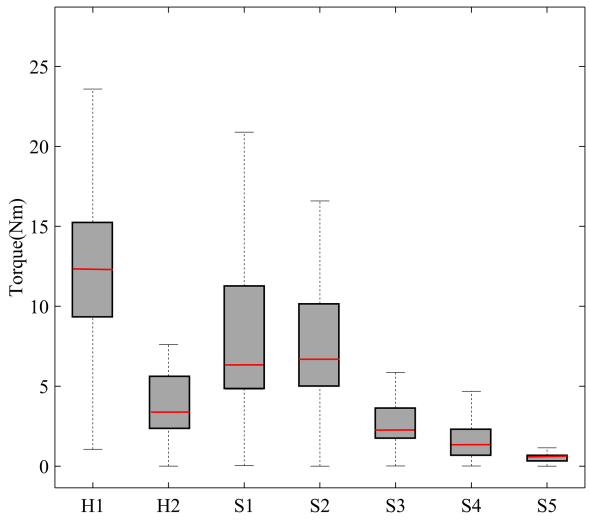


Fig. 7. The box plots show the median (red line), inter-quartile range (box), and minimum and maximum values (tails) of the amplitude of voluntary torque from 2 healthy subjects and 5 stroke survivors.

TABLE V
HUMAN CONTRIBUTION RATIO IN FOUR WORKING MODES OF EACH SUBJECT AND THEIR AVERAGE FROM THE PROPOSED METHOD

CONTROL MODES	HUMAN - DOMINANT			ROBOT – DOMINANT	ALL THE MODES
	ROBOT-RESIST	FREE	ROBOT - ASSIST		
H1	1.04	0.96	0.96	--	1.03
H2	0.90	0.90	0.89	--	0.90
S1	1.01	0.95	0.86	0.72	0.92
S2	1.03	0.94	0.87	0.67	0.92
S3	0.90	0.81	0.69	0.51	0.64
S4	0.78	0.78	0.68	0.36	0.50
S5	0.43	0.39	0.37	0.14	0.22
Average	0.87	0.82	0.76	0.48	0.73

(-- represents did not switch to these working modes for this subject)

S4 and S5, whose P_{hd} were less than 40%. It can be seen that the P_{hd} of subjects decreases with decreasing FMA-LE score.

Fig. 7 and Table IV displayed the ankle joint voluntary torque τ_{ha} of subject. The ankle joint voluntary torque of healthy subjects was basically larger than that of patients. The patients S1 and S2 had large ankle joint torque, which was even larger than that of H2. And $RMS_{\tau_{ha}}$ of subjects decreased as the FMA-LE score decreased except for H2.

The human contribution ratio R_h of two healthy subjects and five stroke patients in four working modes were shown in Table V. The average of R_h decreased in the order of robot-resist mode > free mode > robot-assist mode > robot-dominant mode as listed in Table V. As can be seen in the last column of Table V, R_h of all healthy subjects and most stroke patients were higher than 0.5 during the movement except for S5. R_h of the stroke patients increased with increase in the FMA-LE score. To better demonstrate the advantage of the proposed VAAN controller, our method was compared with a well-recognized EMG-Based AAN controller from [30], which shared the common objective of encouraging patients' voluntary movements with our study

TABLE VI
THE AVERAGE HUMAN CONTRIBUTION RATIO AND AVERAGE TRACKING ERROR IN HUMAN-DOMINANT MODE FROM THIS STUDY AND [30]

METHOD	healthy subjects		patients	
	R_h	RMSE (°)	R_h	RMSE (°)
The proposed method	0.97	2.5	0.76	5.20
Results from [30]	0.74	0.86	--	--

("--" represents that there were no corresponding results in this reference)

in the human dominant mode. The average R_h and RMSEs in human-dominant mode from this study and [30] were shown in Table VI. The results showed that despite a slight bigger tracking errors were appeared, the proposed VAAN controller achieved higher R_h than that of [30].

VI. DISCUSSION

A. Active Participation

Evidence suggested that active participation of patents during rehabilitation training can facilitate neural plasticity and motor recovery [42]. In this paper, we developed a VAAN controller that achieved the purpose of maintaining high-level of patients' active participation during the movement through the following design. Firstly, the voluntary torque of the ankle joint was estimated by an EMG-driven musculoskeletal model to explicitly reflect the movement intention. Secondly, the robot's output torque in human-dominant mode was designed to be proportional to the subject's voluntary torque and the robot is like additional muscles under the voluntary control of the subjects. Although the tracking errors may be affected by the subjects' different levels of disability, this design integrated their voluntary efforts into the control loop and made them realize the significance of their own effort, and thus encouraged them to make active efforts to improve movement performance [43]. In contrast, the relationship between robot assistance of the traditional AAN controllers and patient voluntary torque appeared to be complementary [30], [31], [40], [41], [44]. These controllers will take over most of the physical effort required to complete the movement when subjects exert less effort and still try to keep the trajectory tracking error within a small range even when subjects do not apply any torque [13], [45], [46]. In their studies, improving movement performance was mainly dependent on the assistance from robots, while it was mainly dependent on humans making more efforts in our studies. The increase in R_h demonstrated that the proposed VAAN controller can maintain higher level of active participation than previous studies.

B. Performance-Adaptive Control Strategy

Experiments have demonstrated that both excessive assistance for patients with mild impairment and lack of assistance for patients with severe impairment may inhibit motor recovery [13], [47]. In order to adapt to patients' needs, the proposed VAAN controller can switch among four working modes through adjusting the assistance ratio in this study. The VAAN controller switched to the robot-resist or free mode to increase the training

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difficulty and avoid slackness for patients with mild impairment or healthy subjects (S1, S2, H1, and H2). The R_h value was near 1 in free mode, even greater than 1 in robot-resist mode, which meant that they performed the desired movement mainly on their own effort or even against the robot resistance. Many studies reported that such free or robot-resist mode training can help patients after stroke improve motor function [48], [49], [50]. For patients with severe impairment, the VAAN controller switched to the robot-assist or robot-dominant mode, because training with sufficient assistance may avoid task failure and then patients' frustration, increasing assistance might be necessary to complete the movement task [47]. Unlike [41], [51], [52], where a multi-mode control strategies with fixed control parameters of each working mode was designed for different recovery stages, the proposed VAAN controller can be more adaptive, which can not only match the most suitable working mode according to the level of motor disabilities and the recovery stages of each patient, but also can tailor the robotic assistance to him /her through online adjusting the assistance ratio base on his /her movement performance over time. The assistance ratio was not only applied to adapt to patients' movement performance, but also related to their active participation. Despite the use of EMG-driven musculoskeletal model to estimate patients' voluntary torque and reflect their active participation has been previously suggested in [19], [20], [28], [29], the fixed assistance ratio was adopted in their studies, which meant that the level of active participation of patients cannot adapt to their levels of disabilities and movement performance. In contrast, the proposed VAAN controller dynamically adjusted the assistance ratio according to trajectory tracking error through a forgetting factor, which made a tradeoff between best rendering challenges and improving movement performance to maximize the chance of motor recovery [53].

C. Limitations and Future Work

The limitations that existed in this study can be given in the three aspects. Firstly, some involuntary motor activity of stroke patients, such as spasticity, might contaminate voluntary EMG signals and interfere the effect of the proposed method. The solution to reduce the contamination will be considered in the future. Secondly, both parameters in the control strategy and in the musculoskeletal model may not be optimal in this study. In the future, the control parameters can be optimized represent patients' individualized features and a set of physiologically-based parameters for the EMG-driven musculoskeletal model can be obtained through both static and dynamic calibrations. Finally, although the experiment results verified the feasibility of the proposed approach with a limited number of subjects, the substantial rehabilitation benefits of this approach need to be confirmed through further research on more subjects and long-term training.

VII. Conclusion

This study proposed a voluntary AAN controller for an ankle rehabilitation robot aimed at both adapting to patients' movement performance and promoting their active participation. The voluntary torque of ankle joint was estimated by an EMG-driven musculoskeletal model to reflect patients' active participation and the assistance ratio determined by patients' movement performance has been utilized to switch the control strategies among four working modes. The feasibility of the controller has been validated via experiments with stroke patients and the controller can potentially be applied in clinical rehabilitation.

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